

BoothHub Project

Advanced Booth Booking Management System for Small Businesses

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Project Overview:

BoothHub is a centralized web-based management system designed specifically for small business owners and booth service providers (e.g., coffee booths, photography booths). The project transforms the traditional, manual booking process into a digital, automated workflow to eliminate scheduling conflicts and improve organizational efficiency.

Business Case:

The Problem: Small businesses often manage bookings manually via social media or messaging apps. This leads to disorganized requests, difficulty in tracking customer data, and frequent double-booking errors.

The Solution: A structured administrative portal that allows owners to view real-time statistics, manage their service catalog, and process customer orders through a streamlined interface.

Target Audience: Entrepreneurs in the events and service industry looking for a simple, non-complex management tool.

Technical Specifications:

The system was developed using Core Web Technologies to ensure high performance, stability, and full compatibility across all modern web browsers:

- **HTML5:** Used to build the semantic structure and layout of the dashboard, tables, and administrative forms.
- **CSS:** Employed for modern UI styling. We utilized Flexbox and CSS Grid for responsive layouts, alongside CSS Variables to maintain a consistent "Pastel Color Palette" and professional aesthetic across all pages.
- **JavaScript:** The primary scripting language used to manage application logic, including dynamic data rendering, real-time table updates, and interactive event handling.
- **Web Storage API (LocalStorage):** Implemented for client-side data persistence. This allows the system to store and retrieve booking and service data locally, ensuring information is preserved even after the browser is closed or refreshed.

System Architecture and Functional Description

The BoothHub administrative portal is engineered with a modular architecture where each component is designed to streamline a specific phase of the booking lifecycle. The system moves from secure authentication to real-time data management and strategic decision-making through the following dedicated interfaces:

1. Secure Authentication and Strategic Overview

The administrative workflow begins at the **Login Page**, which serves as the primary security gate for the portal. Developed using a robust JavaScript validation script, this page ensures that sensitive business data is only accessible to authorized personnel. Upon successful credential verification, the system utilizes the `window.location.href` property to facilitate a seamless transition to the Dashboard. As the central command hub, the **Dashboard** provides the administrator with an immediate, high-level summary of the business's performance. Through summary cards, the manager can track real-time statistics such as total booking volume and estimated revenue, allowing for data-driven decisions at a glance.

2. Service Catalog and Inventory Control

To maintain the business's core offerings, the **Manage Services Page** offers a suite of interactive tools for inventory oversight. This interface allows the administrator to dynamically expand the service catalog via the "Add Service" feature, which uses programmatic prompts to capture new service details and render them as interactive cards. Furthermore, the system supports immediate modifications through an "Edit" function and permanent removal via a "Delete" button. This ensures that the user interface is always a direct reflection of the underlying data array, providing a "what-you-see-is-what-you-have" management experience.

3. Dynamic Order Management and Data Processing

The **Manage Orders Page** represents the most technologically advanced component of the system, featuring a highly functional dynamic data table. Designed to manage complex records efficiently, it includes a "New Booking" feature for manual administrative entries. To optimize operational speed, this feature is programmed to automatically assign an "Accepted" status to these entries. The page is further enhanced with sorting algorithms, allowing the manager to organize data by "Highest Amount" or "Newest Date." To maintain a clean and professional UI, a custom "Pagination" logic was implemented, limiting the display to four records per page and utilizing state variables to re-render the table dynamically as the user navigates.

4. Operational Decision-Making and Data Synchronization

The final stage of the administrative process takes place on the **Booking Action Page**, which is dedicated to processing incoming "Pending" requests. Each request is presented as an interactive card, empowering the manager to execute "Accept" or "Reject" decisions. These actions trigger a critical "Data Synchronization" process where the status is updated in the central array and immediately committed to the browser's LocalStorage. This architectural choice ensures absolute data consistency across the platform; any decision made on this page is instantly propagated to the Manage Orders table, ensuring that the system's state is always accurate and up-to-date.

Key Programmatic Logic

1. Dynamic Avatar Generation

We developed a function `getInitials(name)` that splits the customer's name and extracts the first letter of the first and last name.

- Example: "Dr. Nabeel" → DN.

2. Automated Pastel Color Coding

To enhance the visual experience, we implemented a color-coding logic using an array of pastel classes (color-beige, color-pink, color-blue, etc.).

- Logic: `colorClasses[index % colorClasses.length]`. This ensures a diverse distribution of colors across the table rows without manual assignment.

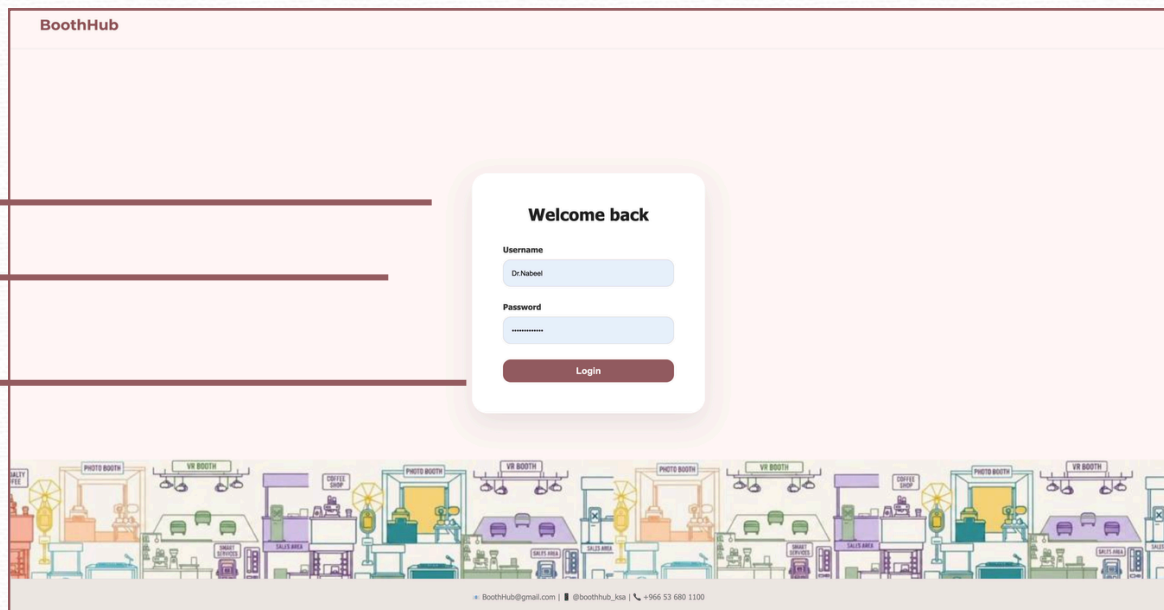
3. Data Persistence

Every change (Add, Delete, or Change Status) triggers a `saveAll()` function.

- Logic: `localStorage.setItem('boothOrders', JSON.stringify(allOrders))`. This converts the JavaScript object into a string for storage, which is then parsed back into an object when the page loads.

BoothHub successfully bridges the gap between manual planning and digital execution. By focusing on a user-friendly interface and robust local data management, the system provides small business owners with the professional tools needed to scale their operations and manage their time more effectively.

1. Login Page



The Login page allows the manager to securely access the BoothHub system using a username and password. The page is designed with a simple and clean interface for easy user experience.

What the Page Displays:

1. Login Form:

The page contains a login form using the `<form>` element with:
Username input, Password input, Login button

2. Login Box Design:

The login form is placed inside the `.login-box` container using:
`box-shadow`, `border-radius`, `display: flex`
to center the form on the page.

3. Background Design:

The page background uses:
`background-image`
with:
`background-position`, `background-size`
to create a professional appearance.

4. Login Button:

The Login button uses the `.btn-login` class with:
`background-color`, `transition`, `cursor: pointer`

5. Login Validation:

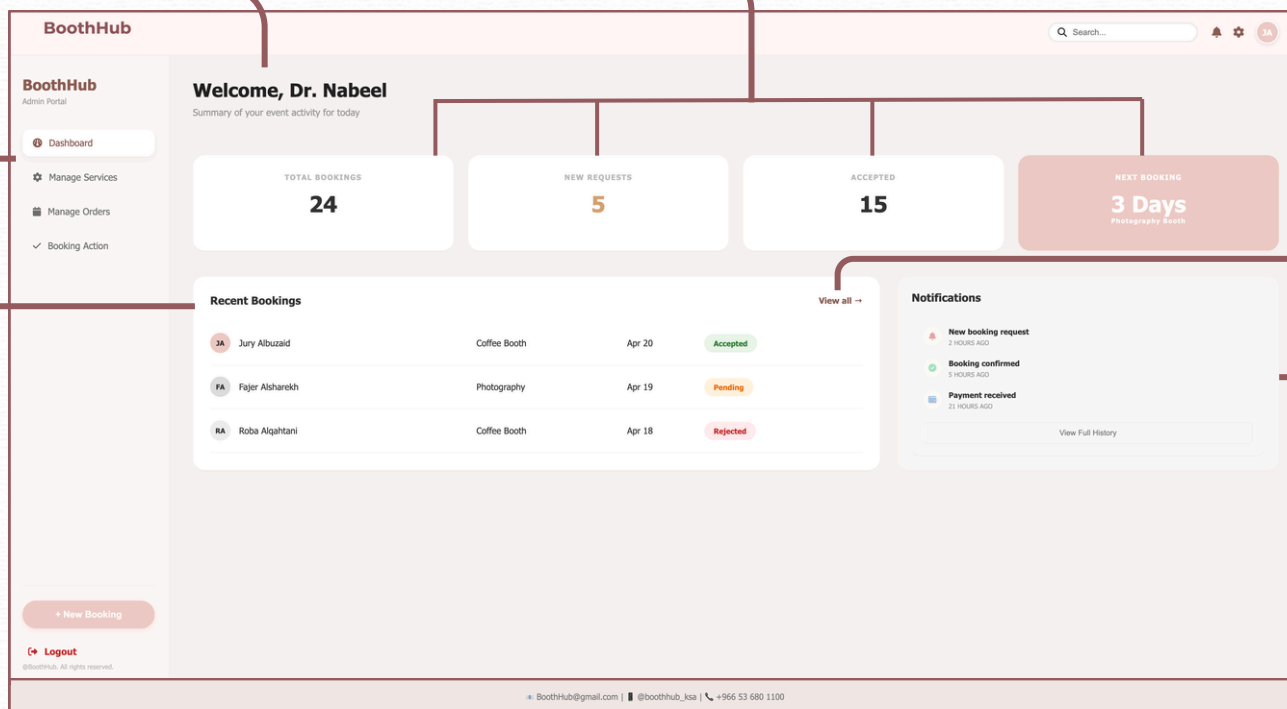
JavaScript is used with:
`addEventListener('submit')`
to check the username and password.

If the login information is correct:
`window.location.href = "dashboard.html";`

The user is redirected to the Dashboard page.

If the information is incorrect:
`alert()`
displays an error message.

2. Dashboard Page

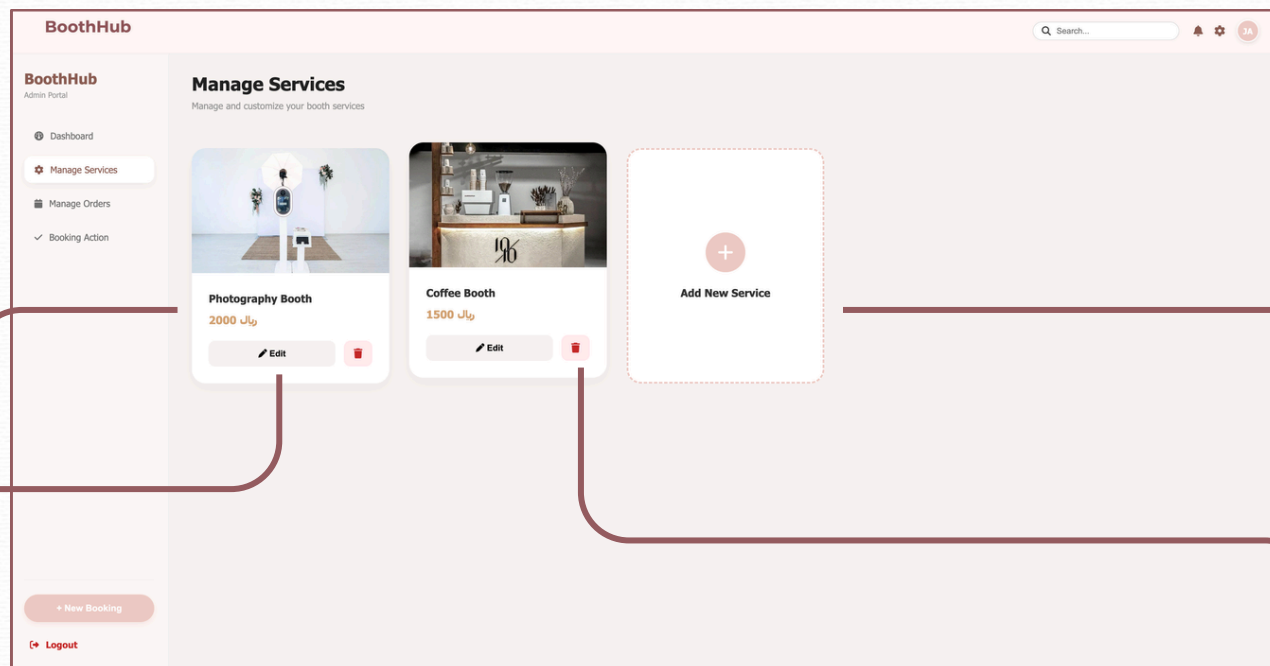


The Dashboard is the main screen of the BoothHub portal. It is designed to be the "control center" where the manager can see everything happening in the business at a glance.

What the Page Displays:

1. **Welcome Header:** At the top, there is a greeting message "Welcome, Dr. Nabeel". It makes the portal feel personal and professional.
2. **Summary Cards:** There are four cards that show the most important numbers:
 - Total Bookings: The overall number of orders.
 - New Requests: Orders waiting for a decision.
 - Accepted: The number of confirmed bookings.
 - Next Booking: A countdown or reminder for the upcoming event.
3. **Recent Bookings Table:** This is the main part of the page. It shows a list of the latest customers, the type of booth they booked, the date, and the status (Accepted, Pending, or Rejected).
4. **Notifications Panel:** This section displays the latest updates and unread alerts. It ensures the manager never misses important activities, such as new booking requests or confirmed payments.
5. **Sidebar & Footer:** On the left, there is a Sidebar for easy navigation between pages. At the bottom, a Footer displays contact information and social media links.
6. **View All Link:** This is a hyperlink located next to the recent bookings table. When clicked, it redirects the user directly to the Manage Orders page to view the complete history of all records.

3. Manage Services Page



The Manage Services page allows the manager to control and organize all booth services available in the BoothHub system. It helps the business owner easily add, edit, update, or remove services from one place in a simple and organized way.

What the Page Displays:

1. Services Cards:

The page displays each service inside a card layout using the `<article>` element with the class `.service-card`. The cards are organized using CSS Grid through the `.services-grid` class. Each card contains:

- Service image
- Service name
- Service price
- Edit button
- Delete button

The card design uses:

- box-shadow for depth effect
- border-radius for smooth corners
- hover effect using transform

2. Add New Service Section:

There is a special card with a plus (+) icon that allows the manager to add a new booth service quickly and easily. This section is created using the `.add-service-box` class

When the manager clicks on the “Add New Service” box, a form appears that allows the user to:

- Add a new service name
- Upload a service image
- Add the service price

3. Edit Service Feature:

Edit

The Edit button allows the manager to update service information such as:

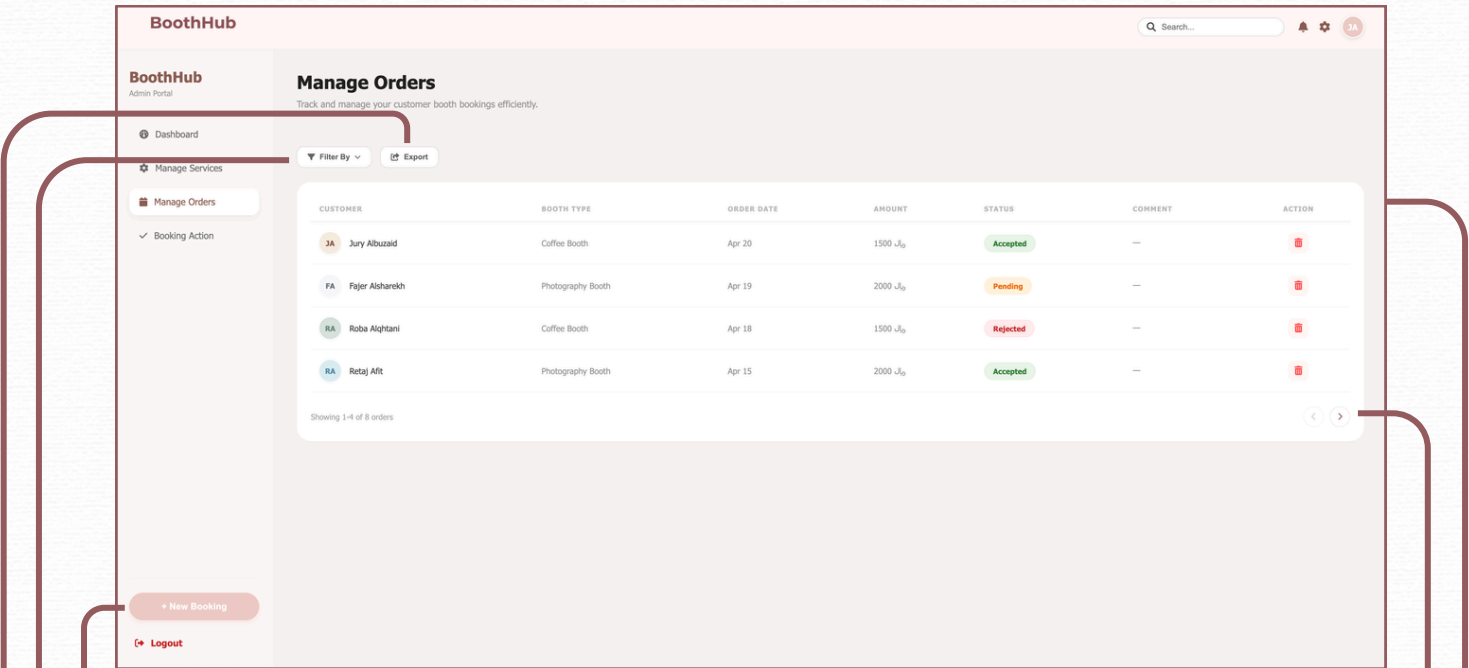
- Booth name
- Price
- Service image

4. Delete Service Feature:



The Delete button allows the manager to remove unavailable or outdated services from the system.

4. Manage Orders Page



The **Manage Orders** page provides a dedicated, interactive dashboard for the system administrator to digitally track, process, and manage all customer booth bookings in an integrated manner. Page Components and Operational Functions:

1. Main Orders Table

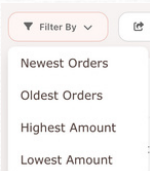
The interface features a dynamic data table that displays active booking records, incorporating the following structural columns: CUSTOMER, BOOTH TYPE, ORDER DATE, AMOUNT, STATUS, COMMENT, ACTION (Row Elimination)

2. Sidebar Action: "New Booking" Portal Input

Located within the persistent administrative sidebar, the "+ New Booking" button allows the system administrator to enter orders manually. When clicked, it initializes sequential input windows (Prompts) via JavaScript to store data across the following fields: Customer Name, Booth Type, Amount (SAR), Date, Comment (Optional) Upon completing this form, the system automatically appends the new entry to the front of the booking matrix.

3. Interactive Filtering Engine ("Filter By")

Data can be filtered using the "Filter By" button. The filtration relies on native array sorting algorithms to reorder transactions based on four primary options: Newest Orders, Oldest Orders, Highest Amount, and Lowest Amount.



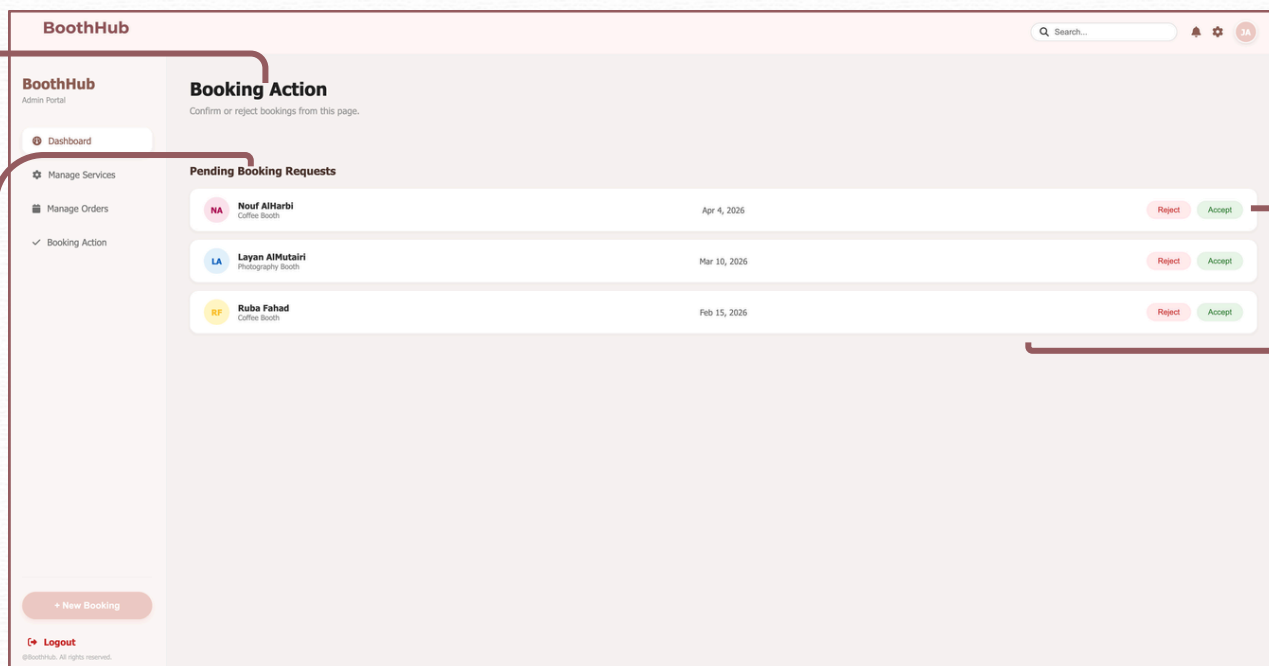
4. Browser Printing Integration ("Export")

The "Export" button hooks directly into the browser using the native JavaScript window printing method window.print(), allowing the system administrator to generate physical printed sheets or save a digital copy.

5. Segmented Navigation (Pagination Control)

The table restricts the viewport to render exactly four records at a time (rowsPerPage = 4) to maintain a professional user experience (UX) design, supported by arrows to navigate between multiple pages.

5. Booking Action Page



The **Booking Action** page allows the manager to review new incoming booth booking requests. The manager can quickly take action by either accepting or rejecting each request.

What the Page Displays:

1. Request Cards:

The page displays each new request inside a clean card layout using the `.request-card` class. Each card contains:

- User avatar and name.
- Booth type.
- Booking date.
- Action buttons (Accept and Reject)

2. Action Buttons Feature:

Accept Button:

Allows the manager to approve the request. When clicked, the request is saved as "Accepted" and moves directly to the Dashboard and Manage Orders tables.

Reject Button:

Allows the manager to decline the request. When clicked, the status changes to "Rejected".

3. Dynamic Card Disappearance:

Once the manager clicks "Accept" or "Reject," the order card immediately disappears to the bottom of the order list to keep the workspace organized and tidy..

6. web structure

